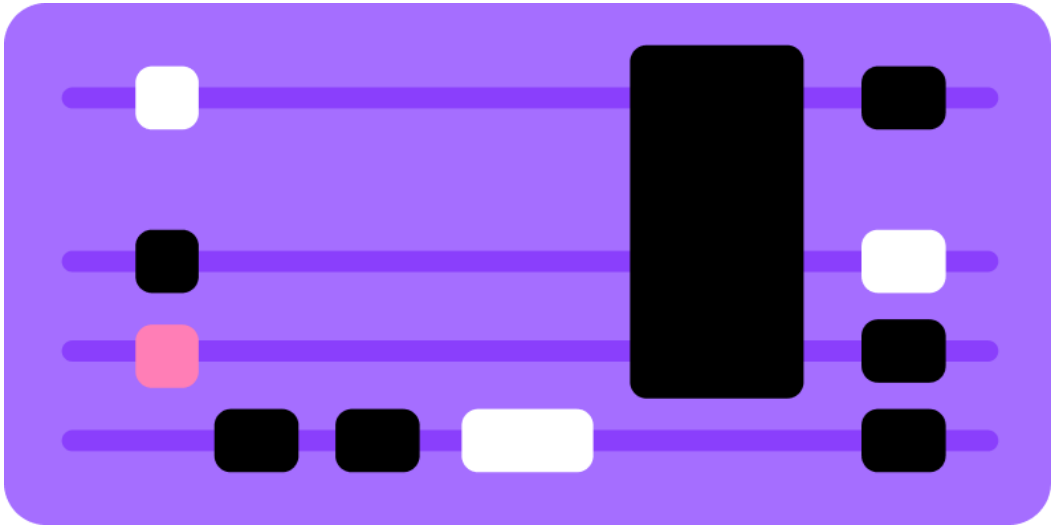
‘Quantum Practitioners’ webinar series

Getting started with Qiskit



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Kifumi Numata

IBM Quantum

kifumi@jp.ibm.com

IBM **Quantum**

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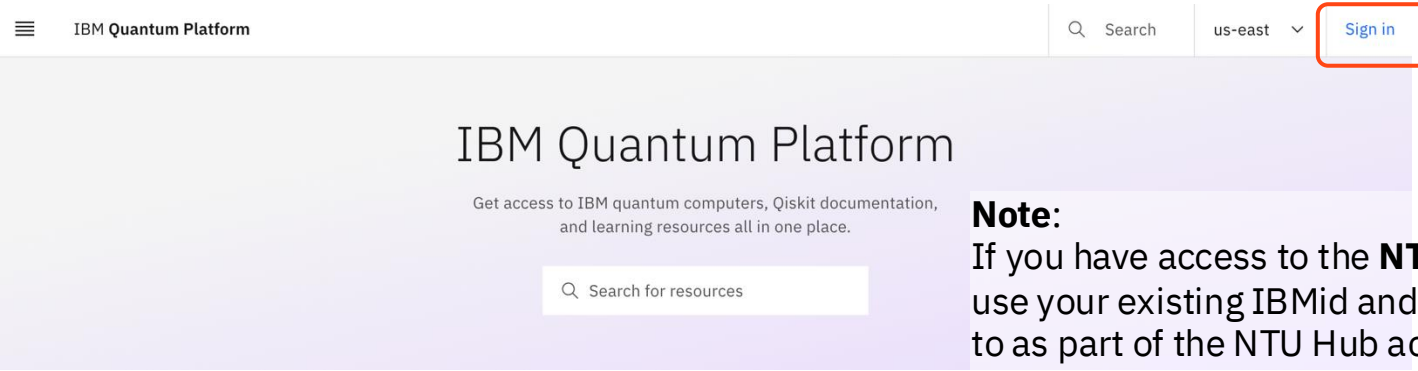
Getting Started with Qiskit

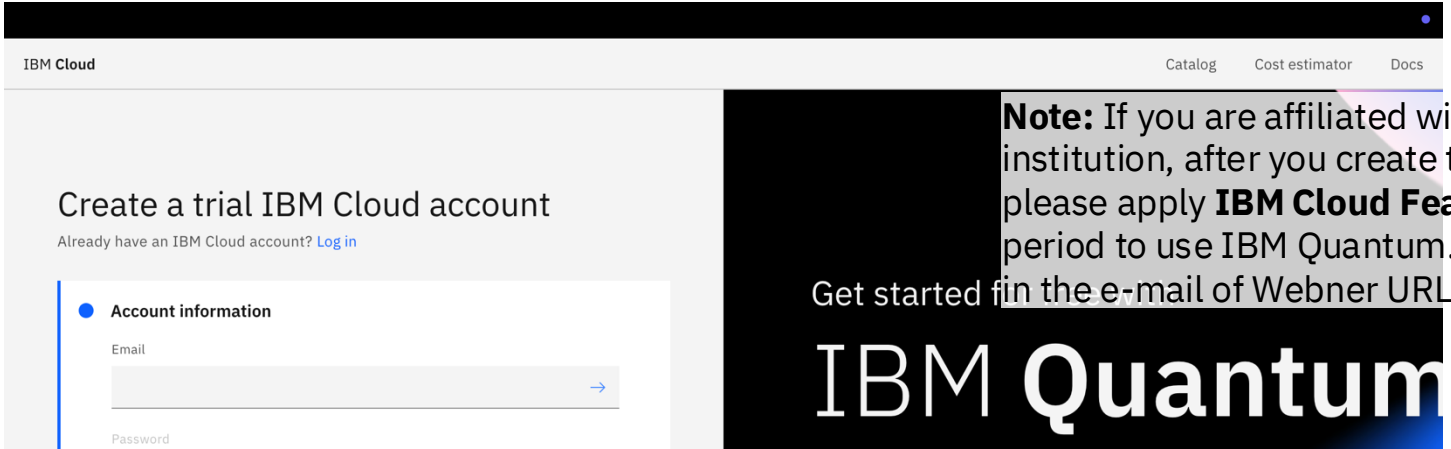
- Qiskit Overview
- Qiskit Install
- Hands-on

Please check if you can sign in https://quantum.cloud.ibm.com/



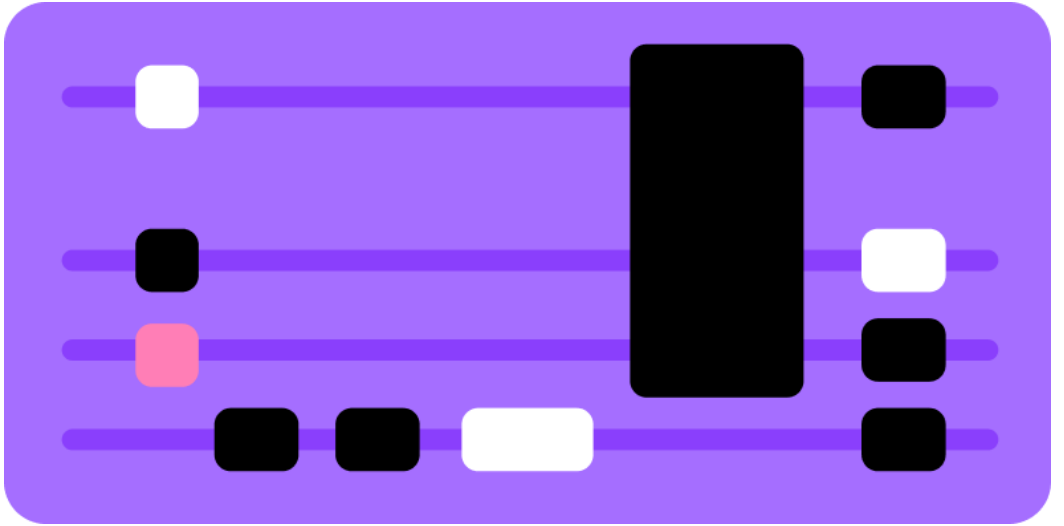
Note:
If you have access to the **NTU Quantum Hub**, please use your existing IBMid and instance you are assigned to as part of the NTU Hub account.

If you can't, please create an IBM Cloud account from https://quantum.cloud.ibm.com/registration



Note: If you are affiliated with an academic institution, after you create the IBM Cloud account, please apply **IBM Cloud Feature code** to extend the period to use IBM Quantum. The guide was included in the e-mail of Webner URL.

Install Qiskit



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Install and set up Qiskit 2.x (macOS)

- Reference URL： https://quantum.cloud.ibm.com/docs/en/guides/install-qiskit (For non-macOS users, please refer this.)
- For those upgrading from version 0.x to 2.x: note that because Qiskit v2.x uses a new packaging structure, you cannot use pip install -U qiskit to upgrade from any Qiskit 0.x version to 2.x.

1. Create a new virtual environment.
   python3 -m venv qiskit-venv
2. Activate the environment.
   source qiskit-venv/bin/activate
3. Install Qiskit and the necessary packages. # zsh users need to put 'qiskit[visualization]' in single quotes.
   pip install qiskit[visualization] qiskit-ibm-runtime jupyter qiskit-aer
4. With the following command, you can launch Jupyter notebook and start using Qiskit.
   jupyter notebook
5. Try the first cell of Quickstart by copy and paste, and execute it by “Shift”+”Enter”.

(When you want to close your qiskit environment, use “deactivate” command to leave it.)
deactivate

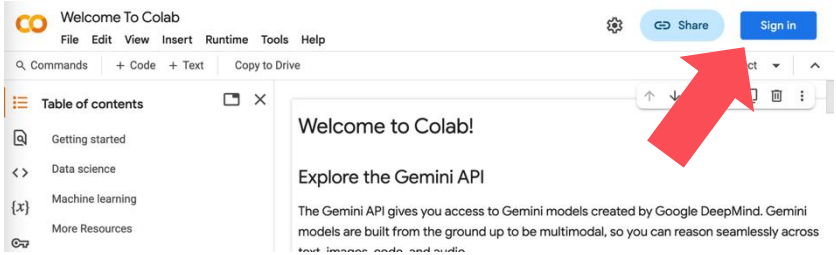
Use Qiskit on Online lab environment

Reference: https://quantum.cloud.ibm.com/docs/en/guides/online-lab-environments

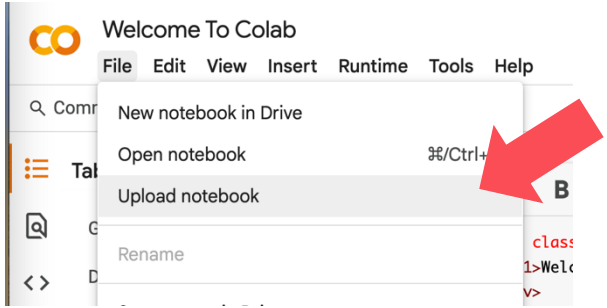
Recommendation is Google Colab because of quick use.

(1) Go to **Google Colab** and Sign in it.

https://colab.research.google.com/



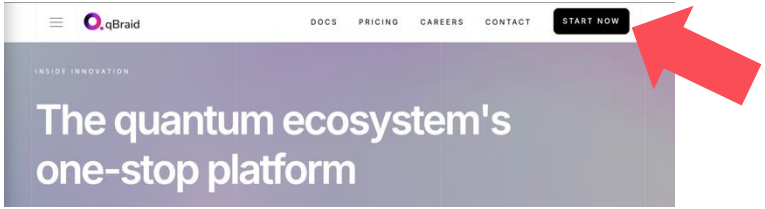
(2) File → Upload notebook



You need to install Qiskit every time when you open new notebook.

(1) Go to **qBraid**.

https://www.qbraid.com/



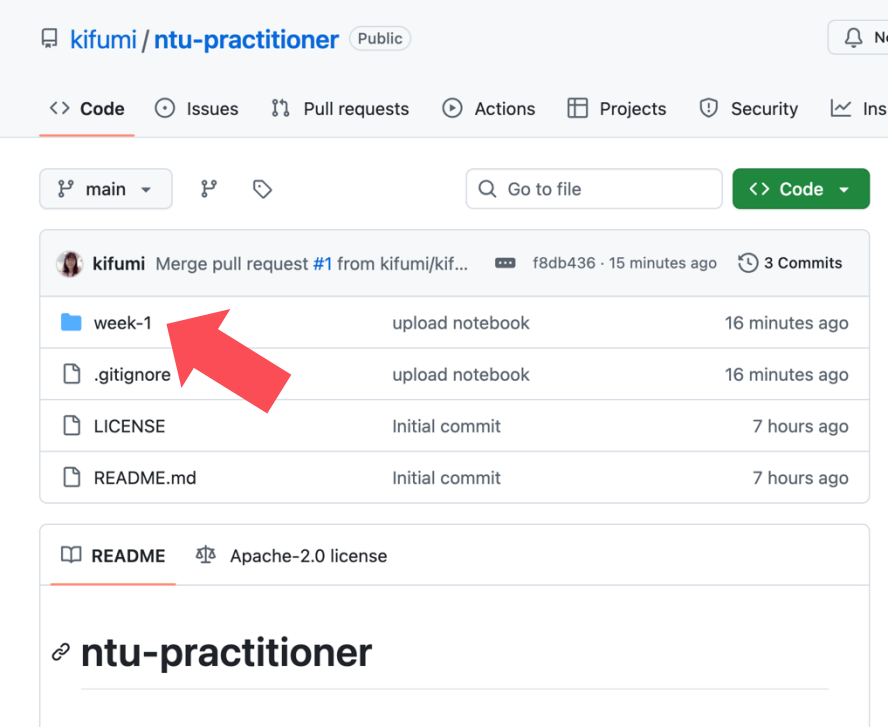
(2) Set up qBraid environment following https://quantum.cloud.ibm.com/docs/en/guides/online-lab-environments#qbraid-lab

You only need to install Qiskit once.

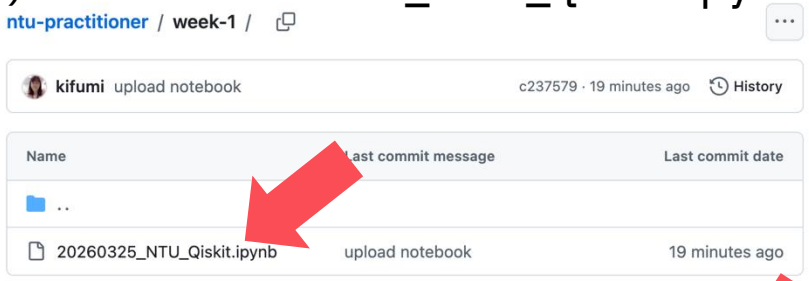
Qiskit hands-on

(1) Go to URL: **ibm.biz/ntu20260325**

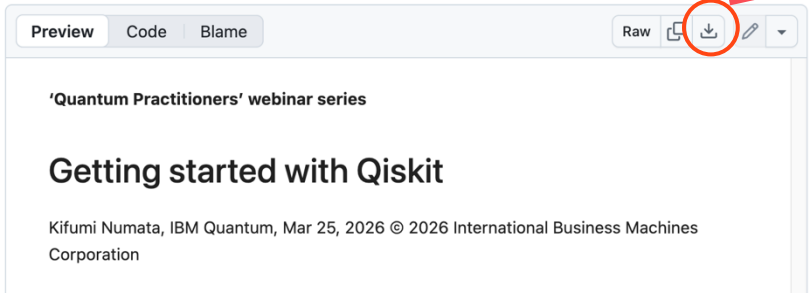
(2) Click “week-1” folder



(3) Click “20260325_NTU_Qiskit.ipynb”



(4) Download raw file



What we learned today

- How to build the quantum circuit using Qiskit
- Qiskit Primitives:
  - Sampler is sampling the output from execution of quantum circuits.
  - Estimator computes the expectation values for the observables with respect to states prepared by quantum circuits.
- How to execute the circuit on real quantum device
- Qiskit Patterns: a standardized design workflow for the practical use of quantum algorithms